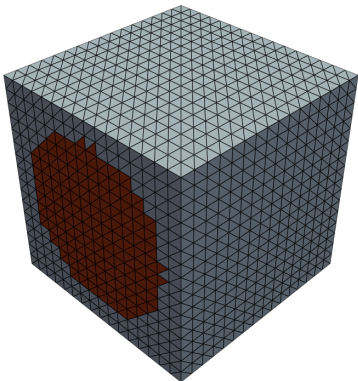
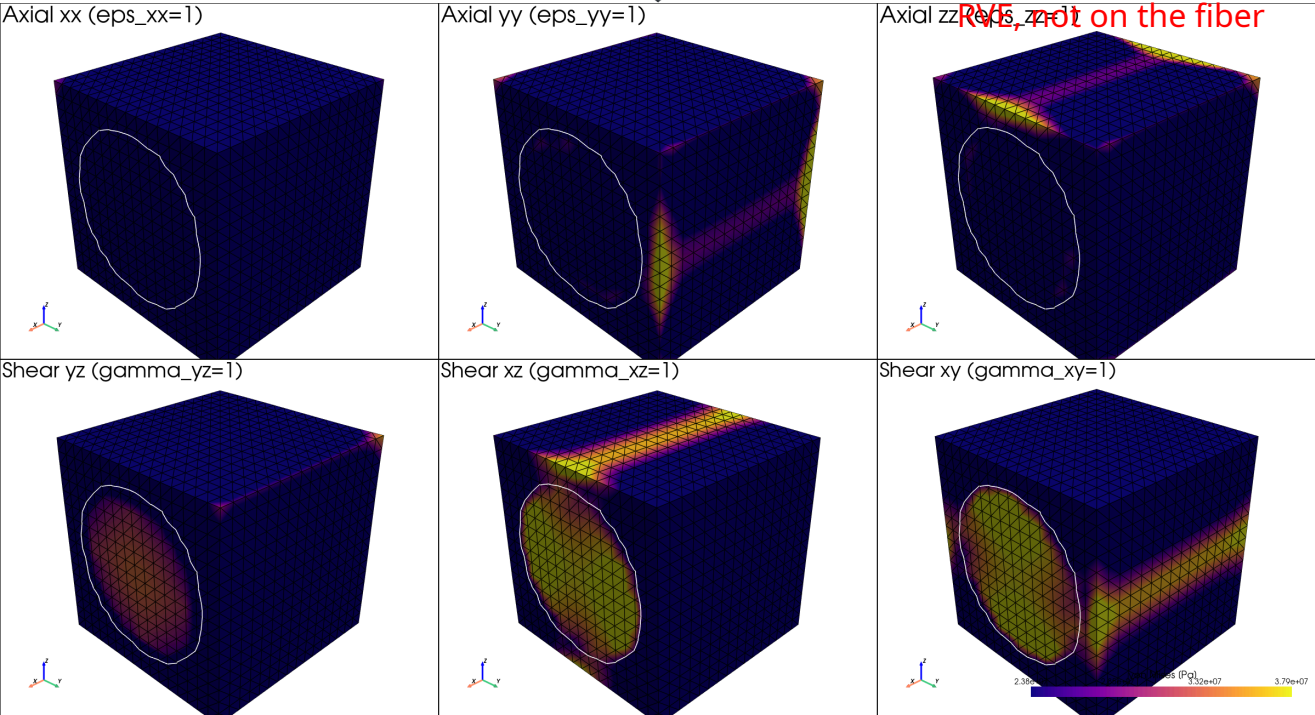


FE mesh (tet) — 16x16x16 structured + 24576 cells; yarn = red, matrix = blue (cell-centroid phase classification)



look at uni1.C
clearly edges are caught
by 2 periodic bcs and corners
by 3
note referring to edges on the
RHS, not on the fiber



INPUTS

material / geometry	values
matrix	E_L=3.0 GPa, E_T=3.0 GPa, G_LT=1.11 GPa, nu_LT=0.35, nu_TT=0.35
yarn	E_L=140.0 GPa, E_T=10.0 GPa, G_LT=5.00 GPa, nu_LT=0.28, nu_TT=0.40
yarn cylinder	r=0.4, vf=0.503, axis=(1.0,0.0,0.0)
mesh	(16, 16, 16) (tetrahedron)

OUTPUTS (effective stiffness)

effective constant	value
E_x, E_y, E_z [GPa]	72.0, 5.9, 5.9
nu_xy, nu_xz, nu_yz	0.311, 0.311, 0.434
G_xy, G_xz, G_yz [GPa]	2.25, 2.25, 1.79
max diag [GPa]	74.12

FULL C_eff [GPa] (Voigt order 11,22,33,23,13,12)

	11	22	33	23	13	12
11	74.12	3.34	3.34	-0.00	-0.00	-0.00
22	3.34	7.42	3.30	-0.00	-0.00	-0.01
33	3.34	3.30	7.42	-0.00	-0.01	-0.00
23	-0.00	-0.00	-0.00	1.79	-0.00	-0.00
13	-0.00	-0.00	-0.01	-0.00	2.25	0.02
12	-0.00	-0.01	-0.00	-0.00	0.02	2.25

BOUNDARY CONDITIONS

where	boundary condition
this figure (visualization)	Periodic: $u(x) = \epsilon_{\text{total}} \cdot x + u_{\text{tilde}}(x)$, u_{tilde} periodic on opposite faces, non-overlapping slave masks per axis (axis 0 excludes $y=L, z=L$ sub-edges; axis 1 excludes $z=L$; axis 2 takes rest), 3 sub-space pins at the geometric centre
homogenization solver (b3-tex solve –backend periodic)	Same periodic BC, 6 unit Voigt strains; effective stiffness = volume-averaged stress per loadcase
KUBC alternative (b3-tex solve –backend kubc)	$u = E_k \cdot x$ on the entire boundary; gives an upper bound for C_{eff}

PERIODIC-BC VERIFICATION (machine-precision residuals over all 6 loadcases)

check	worst residual (across all 6 loadcases)
u_{tilde} slave-master pairs	3.23e-18
u_{tilde} 8 corners agree	0.00e+00
u_{tilde} at centre pin	0.00e+00